

Complex Variables Workshop

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Abstract

This document contains what I intend to cover in the preparatory workshop for the NYU Courant January 2026 complex variables written exams. An emphasis is placed on using key ideas and theorems to solve problems. Thus, most proofs of the main theorems are omitted, unless the proof is particularly enlightening or may appear on the exam. The exact topics covered on each day of the workshop may change depending on how fast/slow we can work through the material.

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Day 1: Review & Differentiation

1.1 Notation and Review

We will define the *complex plane* by

$$\mathbb{C} = \{x + iy : x, y \in \mathbb{R}\} \cong \mathbb{R}^2,$$

where $i^2 = -1$. We write $z \in \mathbb{C}$ to mean that there exists $x, y \in \mathbb{R}$ such that $z = x + iy$. We write $x = \operatorname{Re} z$ and $y = \operatorname{Im} z$, saying that x and y are the *real and imaginary parts* of z , respectively. Addition, multiplication, and division are defined as one would expect,

$$\begin{aligned} (x_1 + iy_1) + (x_2 + iy_2) &= (x_1 + x_2) + i(y_1 + y_2) \\ (x_1 + iy_1)(x_2 + iy_2) &= (x_1x_2 - y_1y_2) + i(x_1y_2 + x_2y_1) \\ \frac{x_1 + iy_1}{x_2 + iy_2} &= \frac{(x_1 + iy_1)(x_2 - iy_2)}{x_2^2 + y_2^2}. \end{aligned}$$

We will denote the *modulus* of a complex number z by the usual Euclidean norm

$$|z| = \sqrt{x^2 + y^2}$$

and *complex conjugation* by

$$\bar{z} = x - iy.$$

It is often convenient to express a complex number in *polar form*, $z = re^{i\theta}$, where $r = |z|$, $\theta = \arg(z)$, and

$$e^{i\theta} = \cos(\theta) + i \sin(\theta).$$

Example 1.1: Find all cube roots of $1 + i$.

Solution. Write $z^3 = 1 + i = \sqrt{2}e^{i(\pi/4+2\pi k)}$, to find that

$$z = 2^{1/6}e^{i\pi/12}, \quad 2^{1/6}e^{i3\pi/4}, \quad 2^{1/6}e^{i17\pi/12}.$$

■

We define the *principal argument* of z , denoted by $\operatorname{Arg}(z)$, to be the unique value in $(-\pi, \pi]$ such that

$$\arg(z) = \operatorname{Arg}(z) + 2\pi n \quad \forall n \in \mathbb{Z}.$$

While it is true that $\arg(z_1 z_2) = \arg(z_1) + \arg(z_2)$, this is not necessarily true for Arg (you should find a counterexample).

Note that the definition of $e^{i\theta}$ immediately gives a definition of the complex exponential function e^z as

$$e^z = e^x e^{iy} = e^x (\cos(y) + i \sin(y)).$$

In these notes, any non-empty, open, connected set D is called a *domain*.

1.2 Analytic Functions

1.2.1 Derivatives of Complex Functions

A function $f: S \subseteq \mathbb{C} \rightarrow \mathbb{C}$ can be decomposed into real component functions u, v via

$$f(z) = f(x + iy) = u(x, y) + iv(x, y).$$

The notions of limits and continuity are defined as one would on a metric space, so we will move right on to discussing differentiation of complex functions.

The derivative of f at a point $z_0 \in \mathbb{C}$ is defined as

$$f'(z_0) = \lim_{z \rightarrow z_0} \frac{f(z) - f(z_0)}{z - z_0}.$$

Taking $\delta z = z - z_0$, this is equivalent to

$$f'(z_0) = \lim_{\delta z \rightarrow 0} \frac{f(z_0 + \delta z) - f(z_0)}{\delta z}.$$

The classical example of a complex function that is not differentiable is the following:

Example 1.2: The conjugation map $z \mapsto \bar{z}$ is not differentiable at the origin.

Solution: Assume instead $f(z) = \bar{z}$ were to be differentiable at $z = 0$. Then the limit of the difference quotient exists on any path. Comparing the paths $(\delta x, 0)$ and $(0, \delta y)$, we find a contradiction. ■

We note that linearity, the Leibnitz rule, the quotient rule, and the chain rule are all true for differentiable complex functions.

1.2.2 Cauchy-Riemann Equations

Perhaps the most important system of partial differential equations (PDEs) in complex analysis are the *Cauchy-Riemann equations*:

Theorem 1.1 (C-R Equations)

Suppose $f = u + iv$ is differentiable at a point $z_0 = x_0 + iy_0$. Then the first order partial derivatives of u, v must exist at (x_0, y_0) , and they satisfy the Cauchy-Riemann equations

$$u_x = v_y, \quad u_y = -v_x.$$

Moreover,

$$f'(z_0) = u_x(x_0, y_0) + iv_x(x_0, y_0).$$

◇

The idea in deriving the C-R equations is to again look at the limit definition of the derivative along the paths $(\delta x, 0)$ and $(0, \delta y)$.

It is important to note that the converse statement is generally not true without additional assumptions on u and v .

Theorem 1.2

Suppose that $f = u + iv$ is defined in some neighbourhood of a point $z_0 = x_0 + iy_0$, and that u, v are continuously differentiable in the neighbourhood and satisfy the C-R equations at (x_0, y_0) . Then $f'(z_0)$ exists, with

$$f'(z_0) = u_x(x_0, y_0) + iv_x(x_0, y_0).$$

◇

This result is the first instance where we see the importance of the local existence of the derivative. We will soon come back to this when we discuss analyticity and integration.

We quickly note that, in polar coordinates, the C-R equations become

$$ru_r = u_\theta, \quad u_\theta = -rv_r.$$

1.2.3 Analyticity

We say that f is *holomorphic* (or *analytic*) at a point z_0 if there exists a neighbourhood of z_0 such that f is differentiable at each point in that neighbourhood. If f is defined on an open set, it is called holomorphic on that set if it is differentiable at each point. A function is called *entire* if it is defined and holomorphic on all of $\mathbb{C}$.

We note that every holomorphic function on a domain D satisfies the C-R equations, and the previous theorems can also be used to prove a function is holomorphic. Another important fact is that if f is holomorphic in a domain D , then the component functions u and v are *harmonic*:

$$\Delta u, \Delta v = 0, \quad \text{where } \Delta = \partial_{xx} + \partial_{yy}.$$

In fact, more can be said, but we first need the following definition. If u, v are harmonic in D and satisfy the C-R equations, then v is called a *harmonic conjugate* of u .

Theorem 1.3

A function $f = u + iv$ is holomorphic in a domain D if and only if v is a harmonic conjugate of u .

◇

Example 1.3: Let $u(x, y) = x^2 - y^2$. Find a harmonic conjugate of u .

Solution: First, note that it is easy to see that u is harmonic. If v is a harmonic conjugate of u , then

$$v_x = -u_y = 2y, \quad v_y = u_x = 2x.$$

Integrating, we see that $v(x, y) = 2xy$ works. ■

Exercise 1.1: Let $u(x, y) = e^x(x \cos(y) - y \sin(y))$.

- (a) Show that u is harmonic on $\mathbb{R}^2$.
- (b) Find a harmonic function v such that $f = u + iv$ is holomorphic on $\mathbb{C}$.
- (c) Express $f(z)$ explicitly as a function of z , where $z = x + iy$.

Example 1.4 (January 2017, Problem 1): Define the function

$$f(z) = \begin{cases} \exp(-\frac{1}{z^4}) & z \neq 0 \\ 0 & z = 0 \end{cases}$$

1. Show that f satisfies the Cauchy-Riemann equations on $\mathbb{C}$.
2. Is f entire?

Solution:

(a) First note that, away from $z = 0$, f is the composition of holomorphic functions. Hence, f satisfies the C-R equations if $z \neq 0$.

Since we are considering the point $(x, y) = (0, 0)$, it suffices to compute the partial derivatives along $x = 0$ and $y = 0$. Decomposing f as $f = u + iv$, it is easy to see that along the axes we have $v \equiv 0$. Meanwhile, $u(x, 0) = \exp(-1/x^4)$, $u(0, y) = \exp(-1/y^4)$. It is a straightforward computation using L'Hopital to show that $u_x(0, 0) = u_y(0, 0) = 0$.

(b) We show f is not continuous at the origin, which proves f is not holomorphic there. To see this, we compute the limit of f as $z \rightarrow 0$ along the line $x = y$. Writing $z = re^{i\pi/4}$, this becomes

$$\lim_{r \rightarrow 0} f(re^{i\pi/4}) = \lim_{r \rightarrow 0} \exp\left(\frac{1}{r^4}\right) = \infty \neq 0 = f(0).$$
■

Example 1.5 (September 2013, Problem 2): Let $z = x + iy$, $f = f(z) = u + iv$. Assume D is a domain in $\mathbb{C}$, $f \in C^2(D)$. Denote

$$Df = \begin{pmatrix} \partial_x u & \partial_y u \\ \partial_x v & \partial_y v \end{pmatrix}.$$

Suppose for every $z \in D$ that

$$Df(z)^T Df(z) = \lambda(z)I,$$

where I is the 2×2 identity matrix. Show that either f or $\bar{f}$ is holomorphic in D .

Solution: Since $Df^T Df = \lambda I$, we have that

$$\begin{aligned}(\partial_x u)^2 + (\partial_y v)^2 &= \lambda = (\partial_y u)^2 + (\partial_x v)^2, \\(\partial_x u)(\partial_y u) + (\partial_x v)(\partial_y v) &= 0.\end{aligned}$$

Note that the second condition is saying that the columns of Df are orthogonal, while the first says that they have the same length (and λ is a non-negative function). Therefore, each column is precisely a rotation of the other by $\pi/2$ or $-\pi/2$.

Let f_1 and f_2 denote the first and second column of Df , respectively. If f_2 is a rotation by $\pi/2$ of f_1 , then

$$\begin{pmatrix} u_y \\ v_y \end{pmatrix} = \begin{pmatrix} -v_x \\ u_x \end{pmatrix}.$$

Thus, the Cauchy-Riemann equations are satisfied, and f is holomorphic.

If instead f_2 is a rotation by $-\pi/2$ of f_1 , a similar argument shows f is antiholomorphic. Therefore, for every point $z \in D$, either f is holomorphic or f is antiholomorphic.

To conclude that only one of these cases can be true on all of D , we need to apply the identity theorem (which we will see later). For now, let us assume the following fact: All zeroes of a non-constant holomorphic function on a connected region are isolated.

Let $J(z) = \det(Df)$ denote the determinant of the Jacobian of f . Since $f \in C^2$, we have that $J \in C^1$. If $z_0 \in D$ is such that f is holomorphic, then by the C-R equations,

$$J(z_0) = u_x v_y - u_y v_x = u_x(u_x) - u_y(-u_y) = u_x^2 + u_y^2 = \lambda(z_0) \geq 0.$$

Likewise, if $z_1 \in D$ is such that f is antiholomorphic,

$$J(z_1) = -(u_x^2 + u_y^2) = -\lambda(z_1) \leq 0.$$

Since D is connected, we can apply the intermediate value theorem to conclude that J attains zero at least once. However, if $\lambda(z^*) = 0$, this implies that

$$u_x, u_y, v_x, v_y = 0.$$

Let us deal with this more precisely. Let

$$\begin{aligned}U &= \{z \in D : J(z) > 0\} \\V &= \{z \in D : J(z) < 0\}.\end{aligned}$$

Note that, by the intermediate value theorem, U and V cannot consist of isolated points. Moreover, by continuity, U and V are open. Define also

$$\Gamma = \{z \in D : J(z) = 0\}$$

If Γ contained an open set, then by analytic continuation f would be constant on D , so we can assume this is not the case. Assume U and V are both non-empty. Since D is connected, one cannot pass from U to V without intersecting Γ .

Let $g(z) = f'(z)$ for all $z \in D$. Then g is holomorphic in U , continuous on $\overline{U}$, and vanishes on $\Gamma \cap \partial U$. If Γ has an accumulation point, then the identity theorem would ensure that g vanishes on all of U , so that f is constant. Thus, Γ can only consist of isolated points. But then $D \setminus \Gamma$ would still be connected, contradicting the intermediate value theorem applied to J (since then the image would be the union of the disjoint intervals $(-\infty, 0)$ and $(0, \infty)$!). Hence, either $U = D$ or $V = D$. ■

Example 1.6: Let $f(z) = u + iv$ be a holomorphic function on some domain $D \subseteq \mathbb{C}$. Assume there exists $\Phi \in C^1(\mathbb{R}^2 \rightarrow \mathbb{R})$ such that

$$\Phi(u(x, y), v(x, y)) = 0 \quad \forall z = x + iy \in D.$$

Assume further that $\nabla \Phi \neq 0$ on $f(D)$. Show that f is a constant.

Solution: Differentiating $\Phi(u, v) = 0$ with respect to x and y and using the C-R equations, we find that

$$\begin{aligned} \frac{\partial \Phi}{\partial u} \frac{\partial u}{\partial x} + \frac{\partial \Phi}{\partial v} \frac{\partial v}{\partial x} &= 0 \\ \frac{\partial \Phi}{\partial u} \frac{\partial u}{\partial y} + \frac{\partial \Phi}{\partial v} \frac{\partial v}{\partial y} &= 0. \end{aligned}$$

This is a system of equations where the determinant of the coefficient matrix is $-(\Phi_u^2 + \Phi_v^2) \neq 0$ on $f(D)$. Thus, this system only has the trivial solution for all $z \in D$, so that

$$f'(z) = u_x + iv_x = 0 \quad \forall z \in D.$$

Hence, f is a constant. ■

1.3 Elementary Functions

Let us make a quick detour to discuss how the exponential, logarithmic, and trigonometric functions generalize to the complex plane.

1.3.1 The complex logarithm

We already defined the complex exponential $e^z = e^x e^{iy}$. However, e^z is periodic with period $2\pi i$, so it is not entirely clear how to define the logarithm such that the logarithm is not multivalued. We define, for $z \neq 0$,

$$\log(z) = \log(|z|) + i \arg(z) = \log(|z|) + i(\text{Arg}(z) + 2\pi k) \quad \forall k \in \mathbb{Z}.$$

Then it follows that $e^{\log(z)} = z$. However,

$$\log(e^z) = z + 2\pi ki \quad \forall k \in \mathbb{Z}.$$

Therefore, we define the *principal value of the logarithm* as

$$\text{Log}(z) = \log(|z|) + i\text{Arg}(z).$$

The function $\text{Log}(z)$ is then well-defined and single-valued for $z \neq 0$. It is straightforward to verify that $\text{Log}(z)$ is continuous on the domain

$$\mathcal{D} = \{z = re^{i\theta} \in \mathbb{C} : r > 0, -\pi < \theta < \pi\}.$$

Moreover, one can check that $\text{Log}(z)$ satisfies the C-R equations on $\mathcal{D}$, so it is also holomorphic. The derivative of $\text{Log}(z)$ is, as one might expect, $1/z$. More generally, $\log(z)$ is analytic on the domain

$$\mathcal{D}_\alpha = \{z = re^{i\theta} \in \mathbb{C} : r > 0, \alpha < \theta < \alpha + 2\pi\}.$$

The line $\theta = \alpha$ is called a *branch cut* of the logarithm, and this definition extends naturally to other multi-valued functions. A single-valued analytic function F that extends to a multi-valued function f is called a *branch* of f . The principal value of the logarithm is often called the principal branch of $\log(z)$.

It is now reasonable to define exponentiation of a complex number by another complex number, via $z^c = e^{c\log(z)}$.

Example 1.7: Express $(1+i)^{-i}$ in the form $x+iy$ for some real x, y .

Solution: We write

$$(1+i)^{-i} = e^{-i\log(1+i)} = e^{-i(\log(\sqrt{2})+i(\pi/4+2\pi k))} = e^{\pi/4+2\pi k} \left(\cos(\log \sqrt{2}) - i \sin(\log \sqrt{2}) \right).$$

■

Exercise 1.2 (September 2002, Problem 1): Find the real and imaginary parts of the complex number $(1+i)^i$.

Exercise 1.3 (September 2005, Problem 1):

- (a) Determine all complex numbers z such that i^z has a finite number of values.
- (b) Same question for i^{i^z} .

1.3.2 Trigonometric & hyperbolic functions

We define the complex sine and cosine functions by

$$\sin(z) = \frac{e^{iz} - e^{-iz}}{2i}, \quad \cos(z) = \frac{e^{iz} + e^{-iz}}{2},$$

and the hyperbolic sine and cosine by

$$\sinh(z) = \frac{e^z - e^{-z}}{2}, \quad \cosh(z) = \frac{e^z + e^{-z}}{2}.$$

A direct consequence of these definitions are the identities

$$\sin(z) = -i \sinh(iz), \quad \cos(z) = \cosh(iz), \quad \sinh(z) = -i \sin(iz), \quad \cosh(z) = \cos(iz).$$

Day 2: Complex Integration

2.1 Contour Integrals

Of particular importance in complex analysis are integrals over closed curves. Before reviewing the key theorems about the integral of a holomorphic function over a closed curve, let us review contour integration.

We define a *curve* C in $\mathbb{C}$ as a continuous function $\gamma: [0, 1] \rightarrow \mathbb{C}$ with $\gamma(t) = x(t) + iy(t)$. The point $\gamma(0)$ is called the *initial point* and $\gamma(1)$ the *terminal point*. If the initial and terminal points are the same, the curve is called *closed*. Note that we could replace $[0, 1]$ in the definition with any real closed interval $[a, b]$ with $a < b$. The curve γ is called *differentiable* if the component functions x and y are continuously differentiable functions of t , with

$$\gamma'(t) = x'(t) + iy'(t),$$

and such that $\gamma'(t) \neq 0$ for all $t \in [0, 1]$. A *contour* is any piecewise differentiable curve. We are mostly interested in *simple curves*, that is, curves that do not intersect themselves. We use the term *positively oriented* to describe a curve that is oriented in the counterclockwise direction.

Given a contour C parametrized by γ on $[a, b]$, we define

$$\int_C f = \int_C f(z) dz := \int_a^b f(\gamma(t)) \gamma'(t) dt.$$

Exercise 2.1: Compute the integral $I = \int_C \bar{z} dz$ when C is given by the curve $z = 2e^{i\theta}$, where $\theta \in [-\pi/2, \pi/2]$.

Example 2.1: Compute the integral $I = \int_C \sqrt{z} dz$ where C is the quarter-circle arc from $z = 1$ to $z = i$ and the integrand is taken with the principal branch.

Solution: The contour C here is parametrized by the curve $\gamma(t) = e^{it}$ for $t \in [0, \pi/2]$. We have

$$I = \int_0^{\pi/2} e^{it/2} (ie^{it}) dt = i \int_0^{\pi/2} e^{i(3t/2)} dt = \frac{2}{3} \left(-\frac{1}{\sqrt{2}} - 1 + i \frac{1}{\sqrt{2}} \right).$$

■

Exercise 2.2: Redo Example 2.1 where the branch is given by $\pi < \theta < 3\pi$. Why is your answer different?

2.2 Analyticity and Integration

In this section, we will demonstrate how analyticity is a much stronger property than mere differentiability. It is likely that a significant portion of the written exam will require the use of the main results of this section, either explicitly or implicitly.

2.2.1 Antiderivatives & Cauchy's Theorem

2.2.2 Important Consequences of Cauchy's Theorem

2.3 More Examples/Exercises

Day 3: Laurent Series & Residues

Day 4: Analytic Continuation & Conformal Mappings